

STA 4102: Computational Methods for Applied statistics

Credit Hours: 3

Department of Mathematics & Statistics

College of Arts & Sciences

COURSE SYLLABUS

Instructor:	Xiankui Yang	Email:	xiankuiyang@usf.edu
Office	CMC 015		
TA/GA	TBA		
Term:	Fall 2024	Dates:	08/26/2024 - 12/12/2024
Lecture:	On Campus TR Location: CMC 209 3:30 PM – 4:45 PM		
Required Textbook:	Not Applicable		
Office Hours:	Instructor: Tuesdays & Thursday from 5:00 PM – 6:00 PM CMC 015 or by email request Office hours start from the week of Aug. 27, 2024. No office hours on holidays.		

I. University Course Description

This course introduces the fundamentals of R statistical software packages, python, Julia, and the SAS (Statistical Analysis Software). Topics include data manipulation, graphs, regression, ANOVA, hypothesis testing, linear models, logistic regression and non-parametric tests using R.

II. Course Prerequisites

STA 2023 (minimum grade C) and STA 3024 (minimum grade C)

III. Course Purpose

This course is about the applications of computational methods in applied statistics using the software, R, which is extensively used in the analysis of the data in both industry and the universities.

IV. Instructor Contact Information and Communication

The best way to reach me is via USF Outlook email. I will respond to your emails within 12 or 24 hours of receiving them. My email is xiankuiyang@usf.edu
Please note that I check my emails a few times a day. It is important that you check the “announcements” in Canvas at least once a day for any communication being posted by me.

V. First Week Attendance Policy

Name call attendance will be taken during the first session on Aug. 27, 2024. Students who cannot make it to the first class must inform me by sending an email no later than 10 PM on the first day.

VI. How to Succeed in this Course

In order to succeed in this course, you are expected attend the classes regularly, run all the practice R programs that will be provided to you as part of my lecture notes to understand the syntax and to interpret the statistical output. Please do not wait until the last minute to understand the practice programs.

VII. Course Structure

The class times are on Tuesday and Thursday, from 3:30 PM – 4:45 PM. The lecture notes along with practice exercises will be provided well in advance of class meetings. You will be assessed by exams, a couple of quizzes/participation and a project.

VIII. Course Topics

- Introduction to data technologies
- Programming fundamentals
- Data structures and variable types
- Sequences and indexing
- Functions, for loops, iteration, simulation
- Plots and charts
- Distributions, CLT, normalization
- Confidence intervals, hypothesis testing
- ANOVA
- Linear Regression
- Logistic Regression.

IX. Student Learning Outcomes

By the end of this course, you will have:

- a) Solid understanding of the syntax of the R programming languages and statistical programming fundamentals
- b) The ability to write complete programs in R to analyze the data
- c) Solid understanding of identifying the procedures (in R) that apply to specific statistical methodology to carry out the analysis of the data
- d) Finally, the ability to make the appropriate interpretation of the statistical output to make inferences on the population data

X. Course Materials

R User Manual: <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>

You must install R-Studio (Posit) on your computer.

Reference Textbook: “Applied Statistics and the SAS Programming Language”, Ed. 5: Ron Cody, Jeffrey K. Smith

Lecture notes will be uploaded in CANVAS as the semester progresses

XI. Grading Categories and Weights

We follow plus/minus grading scale in this course. Below is the template for plus/minus grading:

97 - 100	A+	87 - 89	B+	77 - 79	C+	67 - 69	D+
94 - 96	A	84 - 86	B	74 - 76	C	64 - 66	D
90 - 93	A-	80 - 83	B-	70 - 73	C-	60 - 63	D-
						0 - 59	F

The grade weights are as follows

Assessment	Percent of Final Grade
Exams	40%
Project	30%
Homework	20%
Quizzes/Participation	10%

XII. Instructor Feedback Policy & Grade Dissemination

I will respond to email communication relevant to the subject matter within 24 hours of the date received. Graded tests and materials in this course will be discussed individually only by request. You can access your scores at any time using "Grades" in Canvas..

XIII. USF Standard University Policies

Policies about disability access, religious observances, academic grievances, academic integrity and misconduct, academic continuity, food insecurity, and sexual harassment are governed by a central set of policies that apply to all classes at USF. These may be accessed at:

<https://www.usf.edu/provost/faculty/core-syllabus-policy-statements.aspx>

XIV. Course Policies:

Homework policy:

- There will be **Five** homework assignments.
- All homework will be posted on Canvas with their due dates written on the assignment sheets.
- All homework should be submitted on Canvas by 11:59pm on the due date. No late homework or homework sent via email will be accepted. The lowest score among the six graded homework will be dropped from the final grade calculation

Final Examinations Policy: The final project will be in place of a final exam. The final project is mandatory. Regardless of your performance during the course, failing to complete the project will give you a 'F' grade for the course.

Project Policy:

- All students will complete a term project by the end of the semester.
- The tentative topic of the project is to apply the statistical methods discussed in this class for analyzing real data. Students will find their own data of interest.
- Each student is required to submit a **final project report** as well as giving an **oral presentation** in class. The report should be typed in a word or Latex file. The presentation slides should be made separately (preferred in PowerPoint). The project report is due by **Friday December 06**. No late or hand-written report will be accepted.

Grades of "Incomplete": The current university policy concerning incomplete grades will be followed in this course.

An "I" grade may be awarded to a student only when a small portion of the student's work is incomplete and only when the student is otherwise earning a passing grade. The time limit for

removing the “I” is to be set by the instructor of the course. For undergraduate students, this time limit may not exceed two academic semesters, whether the student is in residence, and/or graduation, whichever comes first.

Make-up Exams Policy: There is no make-up for exams or quizzes or homework or project assignments. But at least speak with me - with proper documentation of dire circumstances make-ups may be considered. There are no curves given for this course

Exam Retention Policy (not applicable for online course): After exams are graded, the instructor will review the examination with the class. The exams will be retained for one semester following the current one, and then they will be destroyed.

Professionalism: Per university policy and classroom etiquette; mobile phones, iPods, etc. must be silenced during all lecture sessions. Those not heeding this rule will be asked to leave the session immediately to not disrupt the learning environment. Students are expected to arrive on time for all class meetings.

Unauthorized Recordings: All unauthorized recordings of class are prohibited. Recordings that accommodate individual student needs must be approved in advance and may be used for personal use during the semester only; redistribution is prohibited.

Early Notification Requirement for Observed Religious Days: Students who anticipate the necessity of being absent from class due to the observation of a major religious observance must provide notice of the date(s) to the instructor, in writing, at the beginning of the term.

Academic Integrity of Students: Instances of academic dishonesty including but not limited to cheating, plagiarism, fabrication, forgery, complicity, and computer misuse will not be tolerated. For further information about university policies for academic dishonesty, go to:
<http://www.ugs.usf.edu/policy/AcademicIntegrityOfStudents.pdf>.

Gender-Based Crimes: USF has a commitment to the safety and well-being of our students. Please be aware that educators must report incidents of sexual harassment and gender-based crimes including sexual assault, stalking, and domestic/relationship violence that come to their attention. I am required to report such incidents for the Office of Student Rights and Responsibilities, or the Office of Diversity, Inclusion, and Equal Opportunity can investigate the incident or situation as a possible violation of the USF Sexual Misconduct/Sexual Harassment Policy and provide assistance to the student making the disclosure. If you disclose in class or to me personally, I must report the disclosure and will assist you in accessing available resources.

Few Emergency Contacts: The Center for Victim Advocacy and Violence Prevention, the Counseling Center and Student Health Services are confidential resources where you can talk about such situations and receive assistance without the incident being reported.

- Center for Victim Advocacy and Violence Prevention: (813) 974-5757.
<http://sa.usf.edu/advocacy>
- Counseling Center: (813) 974-2831.
<http://usf.edu/student-affairs/counseling-center>
- Student Health Services: (813) 974-2331.
<http://usf.edu/student-affairs/student-health-services>.

XV. Course Policies: Student Expectations

Health and Wellness: Your health is a priority at the University of South Florida. We encourage members of our community to look out for each other and to reach out for help if someone is in need. If you or someone you know is in distress, please make a referral at www.usf.edu/sos so that the Student Outreach & Support can contact and provide helpful resources to the student in distress. A 24-hour licensed mental healthcare professional, offered through the counseling center, is available by phone at 813-974-2831, option 3. Please remember that asking for help is a sign of strength. In case of emergency, please dial 9-1-1.

Title IX Policy: Title IX provides federal protections for discrimination based on sex, which includes discrimination based on pregnancy, sexual harassment, and interpersonal violence. In an effort to provide support and equal access, USF has designated all faculty (TA, Adjunct, etc.) as Responsible Employees, who are required to report any disclosures of sexual harassment, sexual violence, relationship violence or stalking. The Title IX Office makes every effort, when safe to do so, to reach out and provide resources and accommodations, and to discuss possible options for resolution. Anyone wishing to make a Title IX report or seeking accommodations may do so online, in person, via phone, or email to the Title IX Office. For information about Title IX or for a full list of resources please visit: <https://www.usf.edu/title-ix/gethelp/resources.aspx>. If you are unsure what to do, please contact Victim Advocacy – a confidential resource that can review all your options – at 813-974-5756 or va@admin.usf.edu.

Generative AI: The purpose of this policy is to foster a dynamic learning environment that encourages technological adaptation, innovative thinking, and the ethical use of AI resources in academic endeavors.

1. **Definition of Generative AI Tools:** Generative AI tools refer to any artificial intelligence-powered software, program or application that can generate content, including but not limited to text, visuals, music, and other creative outputs. Examples of these tools include AI text generators, AI content rewriters, AI graphic generators, etc.
2. **Permitted Use:** The use of generative AI tools is permitted for course-related submissions, including assignments, projects, presentations, examinations, and other forms of assessment. However, students must responsibly use these tools, adhering to the guidelines outlined in this policy.
3. **Student Responsibility:** Students are responsible for appropriately using generative AI tools in their work. This includes:
 - 1) Citing all AI-generated content used in their submissions.
 - 2) Demonstrating a deep understanding of the subject matter, not solely relying on AI-generated content. Cross-reference claims and statements with original sources and providing appropriate citations are expected.
 - 3) Using AI tools as a supplemental resource (i.e., as an editor), not as the primary means of completing assignments.
 - 4) Understanding that generative AI tools, while powerful, are not infallible and can produce misinformation or inaccurate results. Students are responsible for the accuracy of their submissions and must cross-verify the information produced by these tools with reliable sources.
4. **Violation Consequences:** Misuse of AI tools, including use of AI that undermines the student learning objectives of the course or assignment, failing to cite AI-generated content, relying too heavily on AI for work completion, or submitting inaccurate information generated by AI tools, will be subject to academic penalties. Consequences may range from a reduction in an individual assignment grade to larger academic sanctions per USF policy, depending on the severity of the violation ([USF Regulation 3.027](#)).

5. **Exceptions:** If there are specific assignments where the use of AI tools is not appropriate, these will be clearly marked in the assignment guidelines. Students must adhere to these specific instructions.
6. **Questions and Clarifications:** If students are unsure whether a tool they wish to use qualifies as a generative AI tool, or if they have questions regarding the allowable use of such tools, they should consult with the course instructor before using it.

End of Semester Student Evaluations: All classes at USF make use of an online system for students to provide feedback to the University regarding the course. These surveys will be made available at the end of the semester, and the University will notify you by email when the response window opens. Your participation is highly encouraged and valued.

Netiquette Guidelines:

1. Act professionally in the way you communicate. Treat your instructors and peers with respect, the same way you would do in a face-to-face environment. Respect other people's ideas and be constructive when explaining your views about points you may not agree with.
2. Be sensitive. Be respectful and sensitive when sharing your ideas and opinions. There will be people in your class with different linguistic backgrounds, political and religious beliefs or other general differences.
3. Proofread and check spelling. Doing this before sending an email or posting a thread on a discussion board will allow you to make sure your message is clear and thoughtful. Avoid the use of all capital letters, it can be perceived as if you are shouting, and it is more difficult to read.
4. Keep your communications focused and stay on topic. Complete your ideas before changing the subject. By keeping the message on focus you allow the readers to easily get your idea or answers they are looking for.
5. Be clear with your message. Avoid using humor or sarcasm. Since people can't see your expressions or hear your tone of voice, meaning can be misinterpreted.

Food and Drink Policy: Please adhere to the firm policy of no beverages (other than bottled/capped water), food, tobacco products, or like items in the classroom. Your understanding of the necessity for this policy and cooperation will be greatly appreciated. This policy will be strictly enforced.

Email and Discussion Board Guidelines:

1. Use the subject line effectively by using a meaningful line of what your email or discussion is about.
2. Keep your emails and postings related to the course content. You should not post anything personal on a discussion board, unless is requested by the instructor.
3. Any personal, course or confidential issues should be directly communicated to the instructor via email. The discussion boards are public spaces; therefore, any issues should not be posted there.

XVI. Course Technology & Student Support

Academic Accommodations

Students with disabilities are responsible for registering with Student Accessibility Services (SAS) in order to receive academic accommodations. For additional information about academic accommodations and resources, you can visit the SAS website.

[SAS website for the Tampa and Sarasota-Manatee campuses.](#)

Academic Support Services

The USF Office of Student Success coordinates and promotes university-wide efforts to enhance undergraduate and graduate student success. For a comprehensive list of academic support services available to all USF students, please visit the [Office of Student Success website](#).

Canvas Technical Support

If you have technical difficulties in Canvas, you can find access to the Canvas guides and video resources in the “Canvas Help” page on the homepage of your Canvas course. You can also contact the help desk by calling 813-974-1222 in Tampa or emailing help@usf.edu.
[IT website for the Tampa campus](#).

Center for Victim Advocacy

The [Center for Victim Advocacy](#) empowers survivors of crime, violence, or abuse by promoting the restoration of decision making, by advocating for their rights, and by offering support and resources. Contact information is available online.

Counseling Center

The Counseling Center promotes the wellbeing of the campus community by providing culturally sensitive counseling, consultation, prevention, and training that enhances student academic and personal success. Contact information is available online.
[Counseling Center website for the Tampa campus](#).

Tutoring

The Tutoring Hub offers free tutoring in several subjects to USF undergraduates. Appointments are recommended, but not required. For more information, email asctampa@usf.edu.
[Tutoring website for the Tampa campus](#).

Writing Studio

The Writing Studio is a free resource for USF undergraduate and graduate students. At the Writing Studio, a trained writing consultant will work individually with you, at any point in the writing process from brainstorming to editing. Appointments are recommended, but not required. For more information or to make an appointment, email: writingstudio@usf.edu.
[Writing studio website for the Tampa campus](#).

XVII. Important Dates to Remember

Drop/Add Deadline:	Fri, Aug 30, 2024
Labor Day Holiday:	Mon, Sept 2, 2024
Mid-term Grading Opens:	Mon, Oct 7, 2024
Mid-term Grading Closes:	Tues, Oct 22, 2024
Withdrawal Deadline:	Sat, Nov 2, 2024
Veteran's Day Holiday:	Mon, Nov 11, 2024
Thanksgiving Holiday:	Thurs, Nov 28, & Fri, Nov 29, 2024
Final Examination Week:	Sat, Dec 7 - Thurs, Dec 12, 2024

Please check USF academic calendar for more. <http://www.usf.edu/registrar/calendars/>

Disclaimer: This document may have some errors and/or contradicting statements. If you find any, please bring it to my attention. I will rectify/clarify it. *Due to unforeseen circumstances, the instructor reserves the right to make any necessary changes to this syllabus during the semester.* Students will be notified of any changes that are made either via Canvas announcements or class announcements or by both methods. It is the responsibility of the students to make sure to follow these announcements and act accordingly. Sorry – no extra credit. You will receive the grade you have earned so please don't ask for more.